

— DECISION SIMULATION FOR CRISIS LEADERS

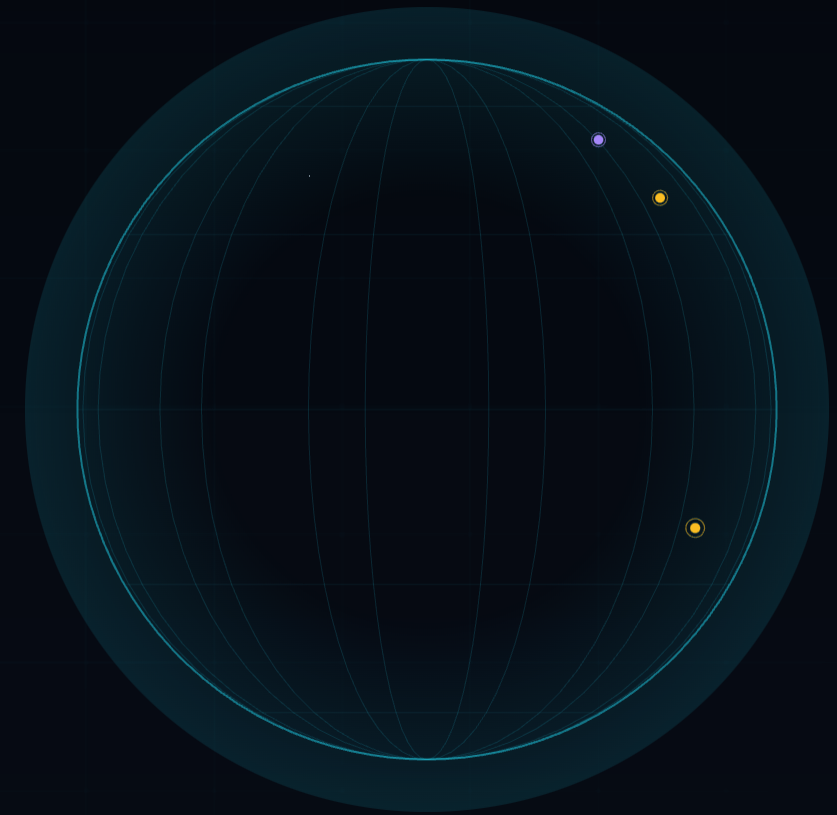
DASHBOARDS SHOW THE CRISIS. WE SIMULATE YOUR DECISION.

A holographic AI war room that ingests live disaster data, forecasts escalation on the GPU, and lays three grounded response options on a 3D Earth — with honest ranges, not false precision.

AMD GPU

ROCM · PYTORCH

PYTORCH MONTE CARLO



— THE GAP

Everyone shows the crisis. Nobody simulates the response.

TODAY — SITUATIONAL AWARENESS

WHAT IS HAPPENING

Crisis24 and Dataminr, public dashboards like GDACS and ReliefWeb — they detect and display events in real time. The feed tells you a flood is rising.

THE MISSING LAYER

WHAT HAPPENS NEXT — UNDER EACH CHOICE

What is exposed in 6 / 24 / 72 hours if nothing is done? Evacuate now, pre-position, or monitor — at what cost, averting whom? **No open-source tool answers this.**

The decision layer is the hard part — the part that needs simulation, grounding, and a GPU. That is exactly what CrisisCommand builds.

— THE DEMO — 60 SECONDS

One globe. Three modes. A decision.

- 01 **Live Earth.** GDACS + USGS events pulse on a holographic globe, severity-colored, ingested minutes ago.
- 02 **Click a crisis.** The camera flies in; an AI situation brief streams in word by word.
- 03 **Run simulation.** 10,000 Monte Carlo runs on the AMD GPU; the readout names the card and spikes; the escalation curve draws itself.
- 04 **Three options.** Evacuate / pre-position / monitor — each with exposed-population range, cost, response time, honest trade-offs.
- 05 **Choose.** Selecting an option renders its evacuation zones and supply arcs on the Earth in 3D.



— WHY IT NEEDS THE GPU — ENGINEERED IN, NOT BOLTED ON

Three compute paths on one AMD GPU.

01 · SELF-HOSTED LLM

SCENARIO AGENT

The three policy branches are one batched LLM request. This demo runs them on **Fireworks** (AMD-powered); the same OpenAI-compatible client points at vLLM on the GPU with one env var — `SIM_BACKEND=vllm`.

02 · MONTE CARLO ENGINE

PYTORCH / ROCM

10,000 stochastic hazard-exposure runs as **one tensor batch**, no Python loops. Device-agnostic — ROCm presents as cuda, so the same code runs on any AMD GPU and on a laptop.

03 · EMBEDDING DEDUP

ON-GPU CLUSTERING

Incoming reports are embedded and cosine-matched on the GPU to merge duplicate events across feeds before they hit the globe.

MONTE CARLO · CPU BASELINE

519 ms · 100k runs

GPU SPEEDUP

61×
100k runs · measured on AMD gfx1100

VRAM ON THE CARD WE RAN

48 GB · gfx1100

GPU READOUT

live
visible during sim

Visible AMD usage is judged AMD usage — the utilization bar sits in the interface and spikes on every run. The speedup is measured on the AMD GPU (61× at 100k runs, 30.4M runs/sec), never estimated.

— THE ETHICAL CORE — THIS IS A DECISION-SUPPORT TOOL

Honest by design. Fake precision is disqualifying.

- ▶ **Every figure is a range.** p10–p90 bands, never a point estimate that pretends to certainty.
- ▶ **The LLM never invents numbers.** The tensor engine quantifies; the model only narrates and applies vetted mitigation factors to the ranges it is handed.
- ▶ **Everything is labeled.** All simulated data carries “SIMULATION — DECISION SUPPORT ONLY.”
- ▶ **A human decides.** Options with explicit trade-offs — never “the AI decided.” Scope is disaster response, never offensive action.

RESILIENCE — REHEARSED, NOT THEORETICAL

- ▶ Dead conference Wi-Fi → full offline demo on curated historical events
- ▶ LLM endpoint down → auto-fallback, honest banner in the HUD
- ▶ Invalid model output → repair retry, then grounded template option
- ▶ Feed rate-limited → per-source freshness dot, other feeds unaffected

Five induced-failure drills pass in CI.

— WHO PAYS FOR THE DECISION LAYER

The buyers already own the dashboards.

GOVERNMENTS

National disaster agencies and emergency managers pre-positioning against a forecast, not a headline.

NGOS

Humanitarian logistics teams choosing where to stage supplies under uncertainty.

INSURERS

Cat-risk desks pricing exposure and modeling loss under intervention scenarios.

LOGISTICS

Supply chains re-routing around unfolding hazards with quantified lead time.

Situational-awareness spend is established and growing. CrisisCommand sells the layer above it — the simulation that turns awareness into a defensible decision.

— BUILD STATUS

Shipped, end to end.

- ▶ **Monte Carlo engine** flood + earthquake kernels, batched tensors, p10–p90
- ▶ **Grounded LLM layer** 3-option generation, briefings, template fallback
- ▶ **Holographic globe UI** Modes A→B→C, escalation chart, zones on Earth
- ▶ **Live ingestion** real GDACS/USGS, on-GPU dedup, freshness
- ▶ **Hardening** quality ladder, 5 failure drills, honest fallbacks

NEXT

- ▶ **AMD GPU deployment** vLLM serving + evidence capture (runbook ready)
- ▶ **Full hazard ensemble** cyclone + wildfire kernels, dynamic weighting
- ▶ **ReliefWeb + tension feeds** richer signal beyond the two solid sources

111 backend tests · smoke test · failure drills — all green.

TEAM IMITASI

COMMERCIAL TOOLS SHOW YOU THE CRISIS. THIS SIMULATES YOUR DECISION.

CrisisCommand – an AI war room, powered by AMD GPUs.

AMD GPU

ROCM

VLLM

FIREWORKS AI

PYTORCH

FASTAPI

REACT · THREE.JS